$$\text{Input} = [t_1, t_2, t_3 \ldots]^{1\times n}$$

$$E = \text{Emb Lookup}[t_1, t_2, t_3 \ldots]^{1\times n} \Longrightarrow E^{n\times d}$$

$$X = \begin{cases} E^{n\times d}\sqrt{d} + \sin\left(\dfrac{w_{pos}}{\delta_i}\right) & \text{if } 2(\text{mod } i) \equiv 0 \quad \text{(even)} \\ E^{n\times d}\sqrt{d} + \cos\left(\dfrac{w_{pos}}{\delta_i}\right) & \text{if } 2(\text{mod } i) = 1 \quad \text{(odd)} \end{cases}$$

$$Z^{(r)} = \text{Attention}(Q, K, V) = \left(\frac{Q_j K_j^T}{\sqrt{d_k}}\right) \odot M_{ij}(-\infty)^{j>i} \cdot V$$

splitting of X

$$\therefore \; Z^{(r)\,n\times d/k} = \text{softmax}\left[\frac{X^{n\times d} W_Q^{d\times d/k} \cdot (X^{n\times d/k} W_K^{d\times d/k})^T}{\sqrt{d_k}} \odot M_{ij}^{n\times n}\right] \cdot X^{n\times d} W_V^{d\times d/k}$$

reintegration

$$\text{MHA}(H)^{n\times d} = \left[Z^{(1)} \,\|\, Z^{(2)} \,\|\, Z^{(3)} \ldots Z^{(k)}\right]^{n\times(k\times d/k) = n\times d} \cdot W_Z^{X\,d\times d}$$

Add back original X and layernorm;

$$x^{n\times d} = LN\left(\text{MHA}(H)^{n\times d} + X^{n\times d}\right)$$

$$= \alpha_x \odot \left[\frac{(\text{MHA}(H) + X) - \mu}{\sigma}\right] + \beta_x \qquad \alpha_x, \beta_x \text{ learnable}$$

<u>Feedforward</u>:

$$FFN(x)^{n\times d} = W_{F2}^{d\times d}\left[\left(x^{n\times d} W_{F1}^{d\times d} + b_1^{1\times d}\right) GELU\right] + b_2^{1\times d}$$

Add back previous step & layernorm

$$O_t^{n\times d} = LN\left(x^{n\times d} + FFN(x)^{n\times d}\right)$$

$$① \text{ —— } O_t^{n\times d} = \alpha_F \odot \left[\frac{(x + FFN(x)) - \mu}{\sigma}\right] + \beta_F$$

$$Y^n = \begin{cases} Embed(Y)^{n\times d}\sqrt{d} + \sin\left(\frac{w_{pos}}{\sigma_i}\right) & \text{if } 2(\text{mod } i) \equiv 0 \quad \text{(even)} \\ Embed(Y)^{n\times d}\sqrt{d} + \cos\left(\frac{w_{pos}}{\sigma_i}\right) & \text{if } 2(\text{mod } i) = 1 \quad \text{(odd)} \end{cases}$$

$$Z^{(r)^{n\times d_k}} = \text{softmax}\left\{\left[\frac{Y^{n\times d} W^{Q^{d\times d_k}} \cdot (Y^{n\times d} W^{K^{d\times d_k}})^T}{\sqrt{d}}\right]^{n\times n} \odot M_{ij}(-\infty)^{j>i}\right\} \cdot Y^{n\times d} W^{V^{d\times d_k}}$$

$$MHA(Y)^{n\times d} = \left[Z^{(1)} \,||\, Z^{(2)} \,||\, \cdots \, Z^{(k)}\right]^{n\times k\times d_k = n\times d} \cdot W_Z^{Y^{d\times d}}$$

Add back original X and layernorm;

$$Y_2^{n\times d} = LN\left(MHA(Y)^{n\times d} + Y^{n\times d}\right)$$

$$= \alpha_Y \odot \left[\frac{(MHA(Y) + Y) - \mu}{\sigma}\right] + \beta_Y$$

$$||$$

$$\Downarrow$$

$\boxed{\text{Cross attention}}$

$$Y_3^{n\times d} = LN\left(Y_2 + MHA_{cross}\left(Q = Y_2,\ K = \overbrace{O_t}^{\text{By } ①},\ V = O_t\right)\right)$$

$$= LN\left(Y_2^{n\times d} + \left[\left\{\left[\frac{Y_2^{n\times d} W_Q^{m\times} \cdot O_t^{n\times d} W_K^{m\times}}{\sqrt{d}} \odot M_{ij}\right] \cdot O_t^{n\times d} W_V^{m\times}\right\} \Big\| \cdots \Big\| \cdots \right]^{n\times h\times d_k = n\times d}\right)$$

$$O_F^{n\times d} = LN\left(Y_3^{n\times d} + \underbrace{\left[W_2^{d\times d}\left[\left(Y_3^{n\times d} \cdot W_1^{d\times d} + b_1^{1\times d}\right) GELU\right] + b_2^{1\times d}\right]}_{FFN}\right)$$

Predict using final output O_F

$$P(t_n \mid t_{n-1}, \ldots)^{n\times v} = \frac{e^{\left(O_F^{n\times d} \cdot W_{out}^{d\times v} + B_{out}^{1\times v}\right)_i}}{\sum e^{\left(O_F^{n\times d} \cdot W_{out}^{d\times v} + B_{out}^{1\times v}\right)}}$$